PES UNIVERSITY

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Campus Placement & Internship Pipeline

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Method	Agile Scrum - two one-week sprint simulations

Prepared from Problem Statement #03 and the Lab 2 Jira Student Handout.

1. Scenario and Requirements Baseline

The proposed system supports the placement cell in managing corporate drives, screening applicants, coordinating multi-round interviews, and processing offers. The primary actors are the **Student Applicant** and the **Placement Officer**.

Functional Requirements

ID	Functional requirement	Priority	Acceptance criteria
FR-001	The system shall accept PDF resumes, extract skills and academic details, and let the student verify the extracted profile.	High	Valid PDF data is extracted and reviewable; invalid or unreadable files receive a clear error.
FR-002	The Placement Officer shall create corporate drives with role, deadline, CGPA, branch and backlog eligibility rules.	High	A published drive displays complete criteria and is visible only during its application window.
FR-003	The system shall evaluate each application against multi-tier eligibility rules and record explicit discrepancy reasons.	High	Eligible candidates enter the pipeline; ineligible candidates cannot bypass filters and see the failed criteria.
FR-004	The system shall maintain ordered queues for multiple interview rounds, schedules, outcomes and applicant notifications.	High	Only cleared candidates advance; round status and schedule remain consistent for both actors.
FR-005	The system shall let officers issue offer letters and let students view, download, accept or decline them.	High	Each response is timestamped, auditable and reflected in the final placement outcome.

Non-Functional Requirements

ID / Type	Description	Priority	Acceptance criteria and rationale
NFR-001 Security & Performance	Encrypt academic records and offer documents at rest with AES-256 and in transit with TLS; enforce role-based access. Eligibility checks should complete within 3 seconds for 95% of requests under expected peak load.	High	Security tests confirm encryption and authorization; load tests meet the target. Sensitive academic and offer data requires confidentiality without delaying drive processing.
NFR-002 Availability & Auditability	Maintain 99.5% monthly availability during placement season and log all rule, round and offer-status changes with actor and timestamp.	Medium	Monitoring confirms availability and audit tests reconstruct a complete change history. Placement decisions must remain reviewable and dependable.

2. Epic Structure

Epic	Purpose	Mapped FRs
EPIC-1 Applicant Profile & Resume	Capture a trustworthy applicant profile from resume and academic data.	FR-001
EPIC-2 Drives & Eligibility	Publish drives, accept applications and apply eligibility rules consistently.	FR-002, FR-003
EPIC-3 Interview Pipeline	Coordinate multi-round queues, schedules, outcomes and notifications.	FR-004
EPIC-4 Offers & Outcomes	Issue secure offers and record student decisions and placement outcomes.	FR-005

The backlog below uses user-centred story statements, explicit business priority and Fibonacci story points. Estimates reflect effort, technical uncertainty and validation risk.

Prioritized Product Backlog

Key / Epic	User story	Pri.	SP	Plan
CPIP-1 / E1	As a student, I want to create and maintain my applicant profile so that companies receive current academic and contact information.	High	3	S1
CPIP-2 / E1	As a student, I want to upload a PDF resume and extract skills so that I do not enter all profile information manually.	High	8	S1
CPIP-3 / E1	As a student, I want to review and correct extracted skills so that my profile accurately represents me.	Medium	3	S1
CPIP-4 / E2	As a placement officer, I want to create a drive with role, deadline and eligibility rules so that applications follow company requirements.	High	5	S1
CPIP-5 / E2	As a student, I want to browse active drives and their criteria so that I can identify relevant opportunities.	High	3	S1
CPIP-6 / E2	As a student, I want to submit an application so that I can participate in an eligible drive.	High	3	S2
CPIP-7 / E2	As a placement officer, I want applications checked against CGPA, branch and backlog rules so that only eligible candidates proceed.	High	8	S2
CPIP-8 / E2	As a student, I want clear discrepancy reasons when ineligible so that I understand which criterion failed.	High	5	S2
CPIP-9 / E3	As a placement officer, I want to configure ordered interview rounds so that candidates advance through the correct process.	High	5	S2
CPIP-10 / E3	As a placement officer, I want to schedule candidates in round queues so that interviews run without conflicts.	High	8	Future
CPIP-11 / E3	As a placement officer, I want to record each round outcome so that qualified candidates advance automatically.	High	3	Future
CPIP-12 / E3	As a student, I want status and schedule notifications so that I do not miss interview actions.	Medium	3	Future
CPIP-13 / E4	As a placement officer, I want to upload and issue an offer letter so that the selected student receives an official document.	High	5	Future
CPIP-14 / E4	As a student, I want to securely view, download, accept or decline an offer so that I can record my decision.	High	3	Future
CPIP-15 / E4	As a placement officer, I want a final outcome dashboard so that accepted, declined and pending offers are visible.	Medium	3	Future

3. Estimation and Sprint Planning

Planning Poker was simulated by comparing three independent estimates and discussing the largest difference before agreeing on the final Fibonacci value. The results below document representative decisions.

Story	Votes	Consensus	Reason
CPIP-2 Resume parsing	5, 8, 8	8	PDF variability, extraction errors and review workflow add uncertainty.
CPIP-4 Drive setup	5, 5, 8	5	Standard form and validation; rules are stored but not yet executed.
CPIP-7 Eligibility engine	5, 8, 13	8	Multiple criteria, failure reasons and bypass prevention require extensive tests.
CPIP-9 Interview rounds	3, 5, 5	5	Ordered transitions and permissions add moderate complexity.

Sprint 1 - Applicant and Drive Foundation (1 week, 22 SP)

Goal: Establish verified applicant profiles and make active drives discoverable.

Sprint 1 - Final simulated board state		
TO DO	**IN PROGRESS**	**DONE**
		CPIP-1 Profile - 3 SP
		CPIP-2 Resume parsing - 8 SP
		CPIP-3 Skill review - 3 SP
		CPIP-4 Drive setup - 5 SP
		CPIP-5 Browse drives - 3 SP

Transition record: Day 2 - CPIP-1 completed; Day 3 - CPIP-2 completed; Day 4 - CPIP-3 and CPIP-4 completed; Day 5 - CPIP-5 completed.

Sprint 2 - Application and Eligibility (1 week, 21 SP)

Goal: Accept applications, evaluate eligibility and prepare interview stages.

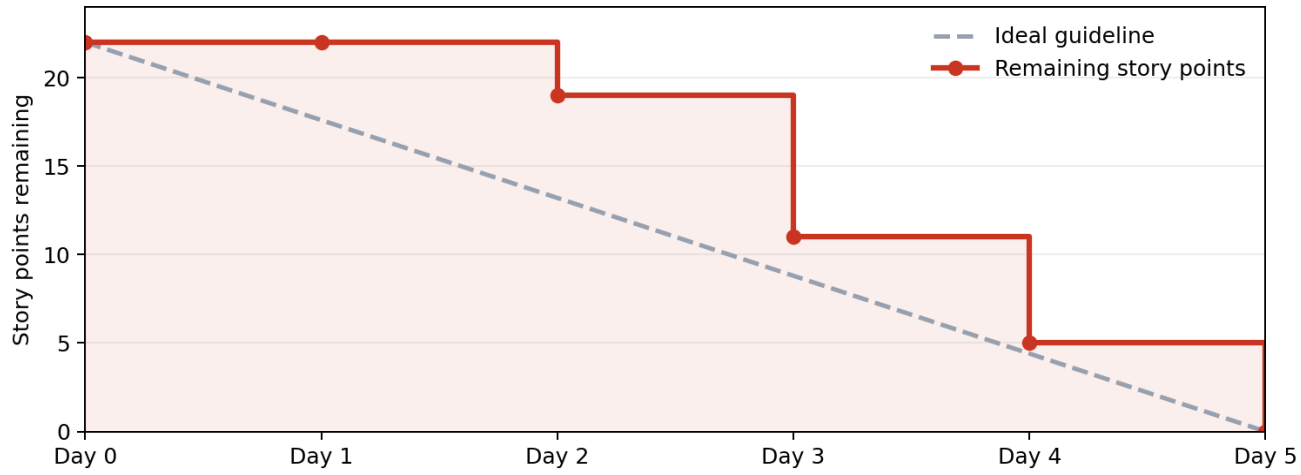
Sprint 2 - Final simulated board state		
TO DO	**IN PROGRESS**	**DONE**
		CPIP-6 Apply - 3 SP
		CPIP-7 Eligibility engine - 8 SP
		CPIP-8 Discrepancy reasons - 5 SP
		CPIP-9 Interview rounds - 5 SP

Transition record: Day 2 - CPIP-6 completed; Day 3 - CPIP-8 completed; Day 4 - CPIP-9 completed; Day 5 - CPIP-7 completed after rule-boundary testing.

4. Burndown Charts

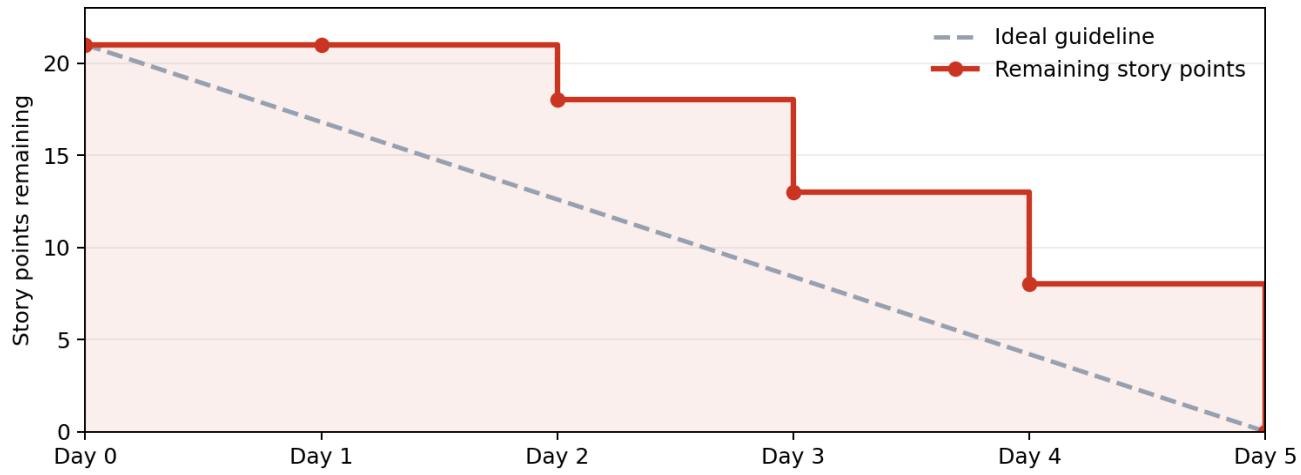
The red step line records remaining story points after each simulated completion. The grey guideline shows ideal linear progress. These charts are reconstructed from the transition record and are not exported Jira screenshots.

Sprint 1 Burndown



Sprint 1 remained at 22 points on Day 1 while setup began, then recovered through steady completions. The 8-point resume-parsing story produced the largest drop. All committed work was completed by Day 5.

Sprint 2 Burndown



Sprint 2 also began slowly, but smaller stories reduced risk before the 8-point eligibility engine was completed. The sprint finished at zero remaining points, indicating that 21 points was achievable in this simulation.

5. Reflection

1. Did your estimations reflect the actual effort?

Mostly yes. The Fibonacci estimates appropriately separated routine profile work from uncertain resume parsing and eligibility logic. CPIP-2 and CPIP-7 required 8 points because input variation, rule combinations and negative testing increased effort. Smaller three-point stories were completed with less uncertainty.

2. Was your backlog well-prioritized?

Yes. High-impact dependencies were placed first: profiles and drive definitions precede applications, while application submission precedes eligibility screening. Interview and offer features remain ordered behind the eligibility foundation. Clear discrepancy feedback was treated as high priority because it directly affects fairness and reduces support queries.

3. How did your simulated sprint align with your plan?

Both sprints achieved their stated goals and completed the committed 22 and 21 story points. Progress was not perfectly linear because larger stories completed as single units, creating step changes. No scope was added during either sprint, so the sprint goals remained stable.

4. What insights did the burndown chart give about your team's capacity?

The simulation suggests a capacity of roughly 21-22 story points per one-week sprint for this backlog. Flat progress at the start indicates that early task decomposition could improve visibility. The sharp drops from eight-point stories show that these items should be split into implementation and validation subtasks in a real project.

Conclusion

The resulting backlog converts the campus placement requirements into four coherent Epics and fifteen user-centred stories. Two simulated sprints deliver the applicant, drive, application and eligibility foundation while leaving interview execution and offer management for subsequent planning.

Evidence status

This report contains a complete Jira-entry specification and transparent sprint simulation. A genuine Jira backlog screenshot, Active Sprint screenshot and Jira-generated burndown screenshot still require an authenticated Jira workspace. They have not been fabricated in this document.